

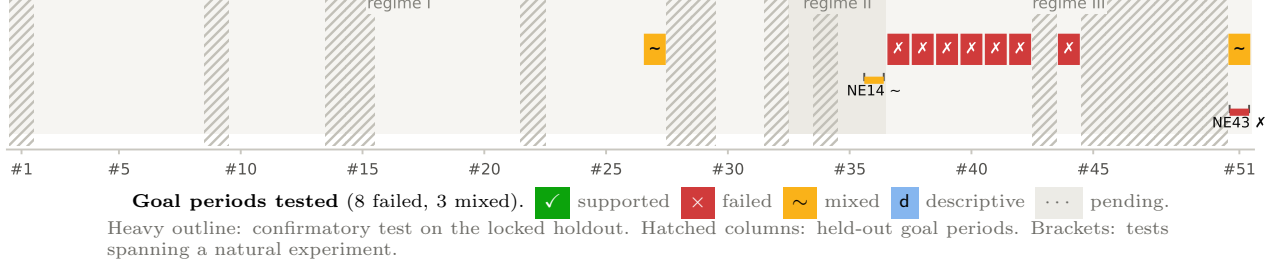
H14: Entropy production of behavior-state sequences: per-family arrows of time, and collective irreversibility

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 1.5/5

Bottom line. Every agent’s fine tool workflow is irreversible even with the scaffold removed, but the coarse arrow and the regime contrast are scaffold structure, and no family, collective or “useful-work” arrow appears.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



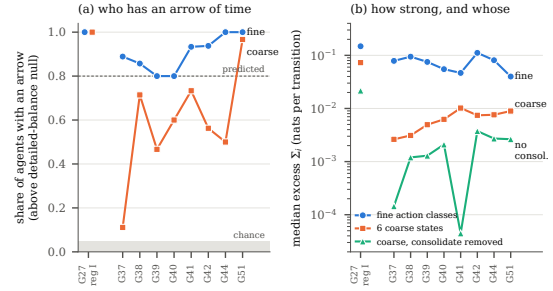
WHAT HAPPENED

Is each agent’s behavior irreversible, does each model family have its own arrow of time (HH19), and is the swarm more irreversible than its agents (HH67)? Round 1 covered 9 periods (#27 in regime I; #37–#44, #51 in regime III). On 6 coarse states arrows are small (0.002–0.010 nats/transition) and mostly the scaffold’s consolidation step; regime I was 7–29× larger; no family or collective term. Round 1b used the improved data: shell sub-classes from the fixed **bash.head**, an agent-only chain with a burn-in after scaffold resets, and Jev v3.1 semantic states. Every agent’s fine arrow is now irreversible (9/9 periods) and survives scaffold removal (41–104% kept); the regime contrast vanishes on every corrected chain; semantic states carry a small pooled arrow (7/9) unrelated to output. Natives (NE43, NE14, loops vs progress) gave no support. Holdout not run.

PHYSICAL MODEL

Kinetic Potts (models 02, 10). $x_i(t) \in \{1..q\}$ is agent i ’s behavior state per turn (single agent) or per 1-min bin (swarm). P_{ab} and π_a are its transition matrix and occupancies. Σ_i is the entropy production per transition: the pair KL, a lower bound for non-Markov agents. It is estimated with the Aguilera–Ito–Kolchinsky dual on antisymmetrized transition indicators g_{ab} , cross-fitted by day (Newton step $\hat{\Sigma}_N$). J_{ij}^{ab} are cross-agent couplings, $\Delta\Sigma$ is the collective increase of the bound, and A_{abc} a cycle affinity.

$$\begin{aligned}\Sigma_i &= \sum_{a,b} \pi_a P_{ab} \ln \frac{\pi_a P_{ab}}{\pi_b P_{ba}}, & g_{ab} &= \mathbf{1}[a \rightarrow b] - \mathbf{1}[b \rightarrow a] \\ \Sigma_g &= \max_{\theta} [\theta \cdot \langle g \rangle - \ln \langle e^{-\theta \cdot g} \rangle], & \hat{\Sigma}_N &= 2 \langle \bar{g}_a^\top K^{-1} \bar{g}_b \rangle_{a \neq b} \\ P(x'_i = a | \mathbf{x}) &\propto e^{h_i^a + \sum_j J_{ij}^{a,x_j}}, & \Delta\Sigma &= \Sigma_{1+2} - \Sigma_1 \\ A_{abc} &= \ln \frac{P_{ab} P_{bc} P_{ca}}{P_{ac} P_{cb} P_{ba}}\end{aligned}$$



Observables. (a) Share of agents whose state sequence beats a detailed-balance surrogate (same occupancies and symmetric transition counts, currents removed). Almost all agents do on fine action classes; on the 6 coarse states only a minority do, except in the long #51 window and in regime I (#27). (b) Median excess irreversibility. Fine micro-workflow carries about 10× the coarse arrow, and most of the coarse arrow vanishes when the scaffold’s consolidation step is removed.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Mostly a careful measurement of the harness, now with one agent-level residue. The coarse arrow and the regime-I/III contrast are scaffold structure (sessions, consolidation). What survives removing the scaffold is fine tool workflow (e.g. read → edit → commit), irreversible in every agent; it is a per-agent fingerprint, not a steering handle. Semantic irreversibility is real but small and does not track output or progress, so EP is not a “useful work” meter. Practical lessons: use a calibrated detailed-balance or block-flip null, null-subtract before comparing agents, and never apply a reset burn-in on coarse windows (it manufactures arrows). An operator would change nothing.

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RESULTS VS PREDICTIONS

prediction (round 1b)	observed	verdict
fine (shell sub-classes)	100% of agents in 9/9	holds
$\geq 80\%$ irreversible		
coarse arrow mostly	5/8 keep $< 50\%$ after removal	failed (narrow)
scaffold ($\geq 6/8$)		
fine arrow survives	kept 41–104%; 75–100% of holds	
scaffold removal	agents	
pooled Jev v3 arrow	$p \leq 0.005$; 7/9 periods	holds
(G38, G51)		
regime I $\geq 2\times$	(v3 ratios 0.4–4.4)	failed
states)		
lab η^2 (fine, scaffold-free)	p 0.35; Anthropic > OpenAI	failed / fragile
	8/8, p 0.04	
v3 EP tracks output	ρ +0.14, -0.34, -0.16	failed
N1 NE43 inside	3/4 changes inside	failed (marg.)
placebo range		
N2 NE14: coarse	0.25 vs 0.72 (CIs span 1)	mixed
drops more than v3		
N3 productive > stuck	0/3	failed
EP		

GOAL PERIODS IN DETAIL

G27	$\sim 1b$: mixed (unchanged: the pre-registered coarse chain...
G37	$\times 1b$: failed (unchanged: the pre-registered coarse chain...
G38	$\times 1b$: failed (unchanged: the pre-registered coarse chain...
G39	$\times 1b$: failed (unchanged: the pre-registered coarse chain...
G40	$\times 1b$: failed (unchanged: the pre-registered coarse chain...
G41	$\times 1b$: failed (unchanged: the pre-registered coarse chain...
G42	$\times 1b$: failed (unchanged: the pre-registered coarse chain...
G44	$\times 1b$: failed (unchanged: the pre-registered coarse chain...
G51	$\sim 1b$: mixed (unchanged: the pre-registered coarse chain...
NE14	$\sim$ mixed
	+1 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

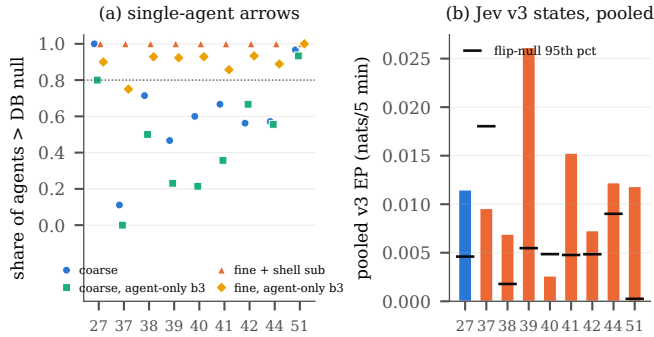
Soft-label observables attenuate EP roughly by the square of label accuracy, so per-agent v3 tests are nearly blind and only pooled tests are primary. A reset burn-in on 5-min windows was pre-registered and turned out to be a selection artifact (resets start activity bursts); it is reported as invalid. Newton blows up for a few strongly asymmetric agents (one G40 value 16.5), which drives the fragile Anthropic > OpenAI result. NE14 has one pre-change day; NE43 windows are 5 days. Non-Markov agents make all values lower bounds.

ROUND 2

Count-matched within-agent EP across goal kickoffs (H56's design) on `act.sh`; HMM-denoised semantic states before EP.

- **Essence.** The arrow of time of work: progress is irreversible, chatter is reversible.
- **H14-R1** Entropy production on work states (plan, build, verify, ship; Jev) correlates with output: useful irreversibility.
- **H14-R2** Loops are near-reversible cycles with zero progress; productive phases produce entropy.
- **H14-R3** The scaffold contributes a fixed entropy-production floor (the consolidation clock); subtracting it leaves agent-generated irreversibility (E1).

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Second figure. (a) Share of agents beating the detailed-balance null per period. Fine classes with shell sub-classes (triangles) are irreversible for everyone and stay so with the scaffold removed (diamonds); the coarse arrow (dots) mostly does not survive removal (squares). (b) Pooled Jev v3 entropy production per 5-min transition (bars) and the block-flip null's 95th percentile: regime I (#27, blue) is not above regime III.